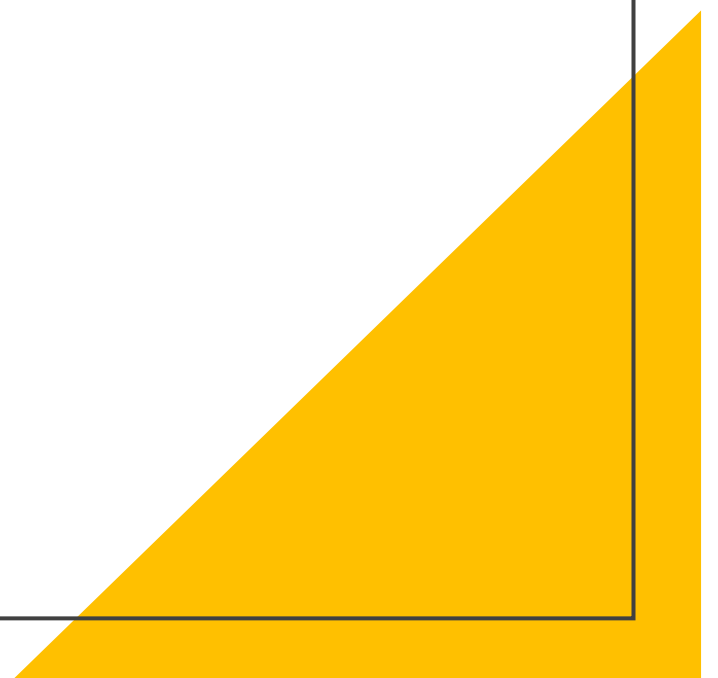


Design Science Research

Dissertation (EI or ETI)

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Summary



What is Design Science Research (DSR)



Design and Behavioural Science



Design



Models of DSR



Guidelines for DSR

Design Science Research

- ❑ Roots in engineering and the sciences of the artificial (Simon 1996)
- ❑ Problem solving paradigm.
- ❑ Seeks to create innovations that define the ideas, practices, technical capabilities, and products through which the analysis, design, implementation, management, and use of information systems can be effectively and efficiently accomplished (Hevner et. al, 2004; Denning 1997;Tsichritzis 1998).

Why?

- **Design Science** (DS) the scientific study of the artificial, as opposed to the natural (Simon, 1996)
- Defs. Of Science (Simon 1996):
 - ❑ **Natural Science:** research in physics, biology, economics and sociology. Concerned with explaining reality (how, and why things are)
 - ❑ **Design Science:** problem-solving paradigm that extends the boundaries of human and organisational capabilities by creating new and innovative artefacts.

Design Science and Behavioural Science

- ❑ **Design-science paradigm** seeks to extend the boundaries of human and organizational capabilities by creating new and innovative *artifacts*.
- ❑ **Behavioural science paradigm** seeks to develop and verify theories that explain or predict human or organizational behaviour.



Design Science and Behavioural Science

❑ From (Winter, 2008):

- ❑ **Behavioural Research:** aims at 'truth', i.e., at the exploration and validation of generic cause-effect relations
- ❑ **Design Science Research:** aims at 'utility', i.e., at the construction and evaluation of generic means-end relations

What implies...?

- ❑ Design Research complies:
 - ❑ 2 design processses:
 - ❑ **Build**
 - ❑ **Evaluate**
 - ❑ 4 artifacts:
 - ❑ **Constructs:** provide the language in which problems and solutions are defined and communicated (Schön 1983).
 - ❑ **Models:** use constructs to represent a real-world situation. The design problem and its solution space (Simon 1996).

What implies...?

- ❑ Design Research complies:
 - ❑ **4 artifacts (cont.):**
 - ❑ **Methods:** define processes. They provide guidance on how to solve problems, that is, how to search the solution space.
 - These can range from formal, mathematical algorithms that explicitly define the search process to informal, textual descriptions of *best practice* approaches, or some combination.
 - ❑ **Instantiations:** show that constructs, models, or methods can be implemented in a working system.
 - They demonstrate feasibility, enabling concrete assessment of an artifact, suitability to its intended purpose. They also enable researchers to learn about the real world, how the artifact affects it, and how users appropriate it.

- ❑ **Build → evaluate → theorize → justify** (March and Smith, 1995, p.258).
 - ❑ Building: process of constructing an artefact for a specific purpose, demonstrating that such as artefact can be constructed.
 - ❑ Evaluation refers to the development of criteria (measures) and assessment of artefact performance against those criteria. The evaluation of the artefact provides feedback information and a better understanding of the problem domain that is used to improve both the quality of the product and the design process.
 - ❑ This build and evaluate loop is typically repeated several times to achieve the final design artefact (Hevner et al., 2004, p.78).
- ❑ **Identify a need → Build → Evaluate → Learn → Theorize** (Rossi and Sein, 2003).

Reference Models for Design Science Research

❑ **Develop/build → justify/evaluate**
(Hevner et al., 2004, p.79).

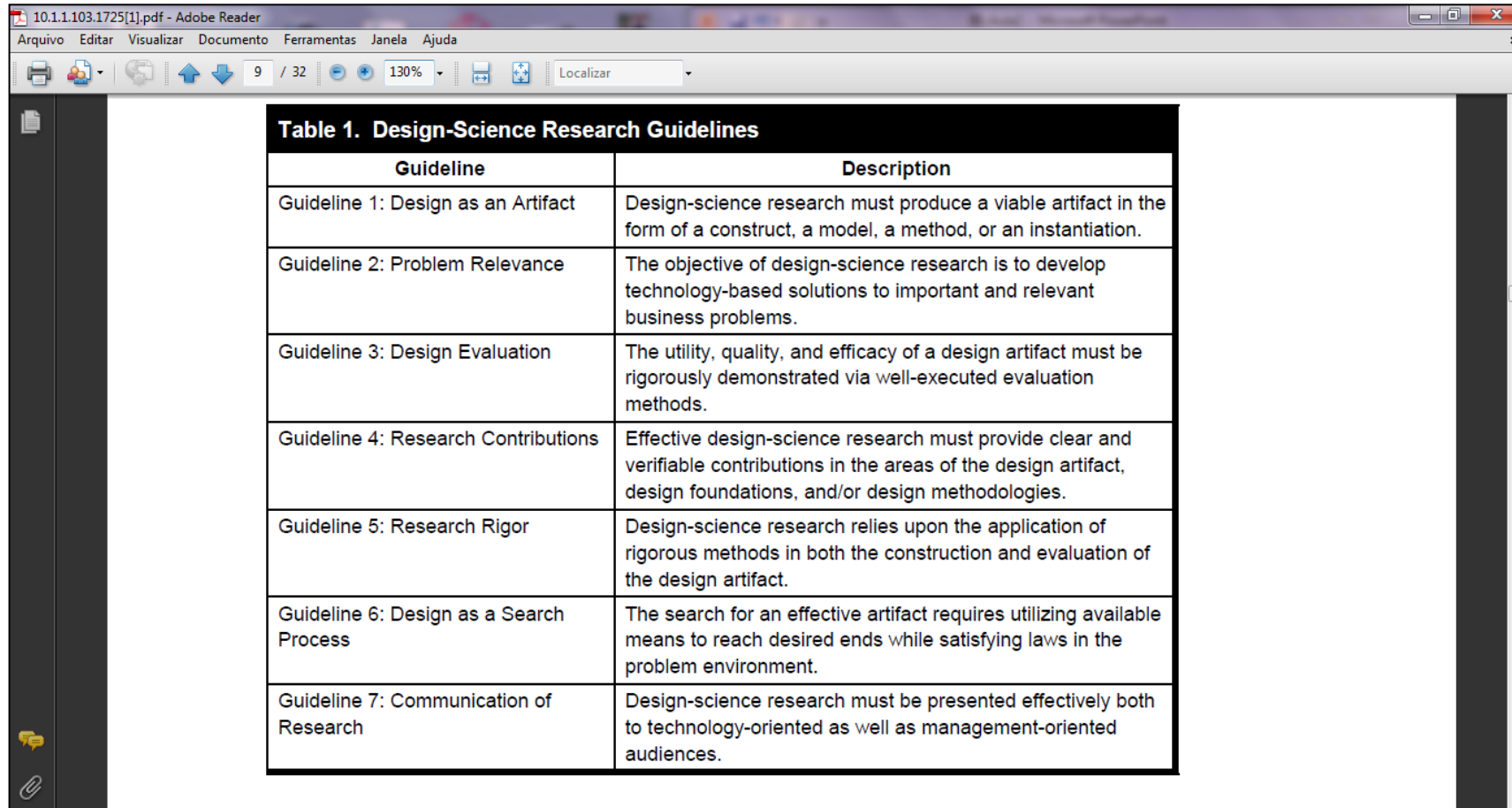
❑ **Problem identification and motivation →
Objectives of a solution → Design and
development → Demonstration →
Evaluation → Communication** (Peppers et
al., 2006, Peppers et al., 2007-8)

Reference Models for Design Science Research

Guidelines what for ...?

- ❑ **Design Science Research** is inherently a problem-solving process
- ❑ Fundamental Principle:
 - ❑ Knowledge and understanding of a design problem and its solution is acquired in the building and application of an artifact
- ❑ Design Science Research requires:
 - ❑ **Creation of an innovative purposeful artifact** (G.1.)
 - ❑ For a **specified domain** (G.2.) yielding utility for the **specific problem**
 - ❑ **Evaluation of the artifact's utility** (G.3.)
 - ❑ **Innovation** (G.4.): artifact must solve a heretofore unsolved problem or solving a known problem in a more effective or efficient manner
 - ❑ **Artifact must be rigorously defined**, formally represented, coherent and internally consistent (G.5.)
 - ❑ **Problem space identified** and a mechanism posed or enacted to find an effective solution (G.6.)
 - ❑ **Results must be communicated** (G.7.)

Design Science Research Guidelines



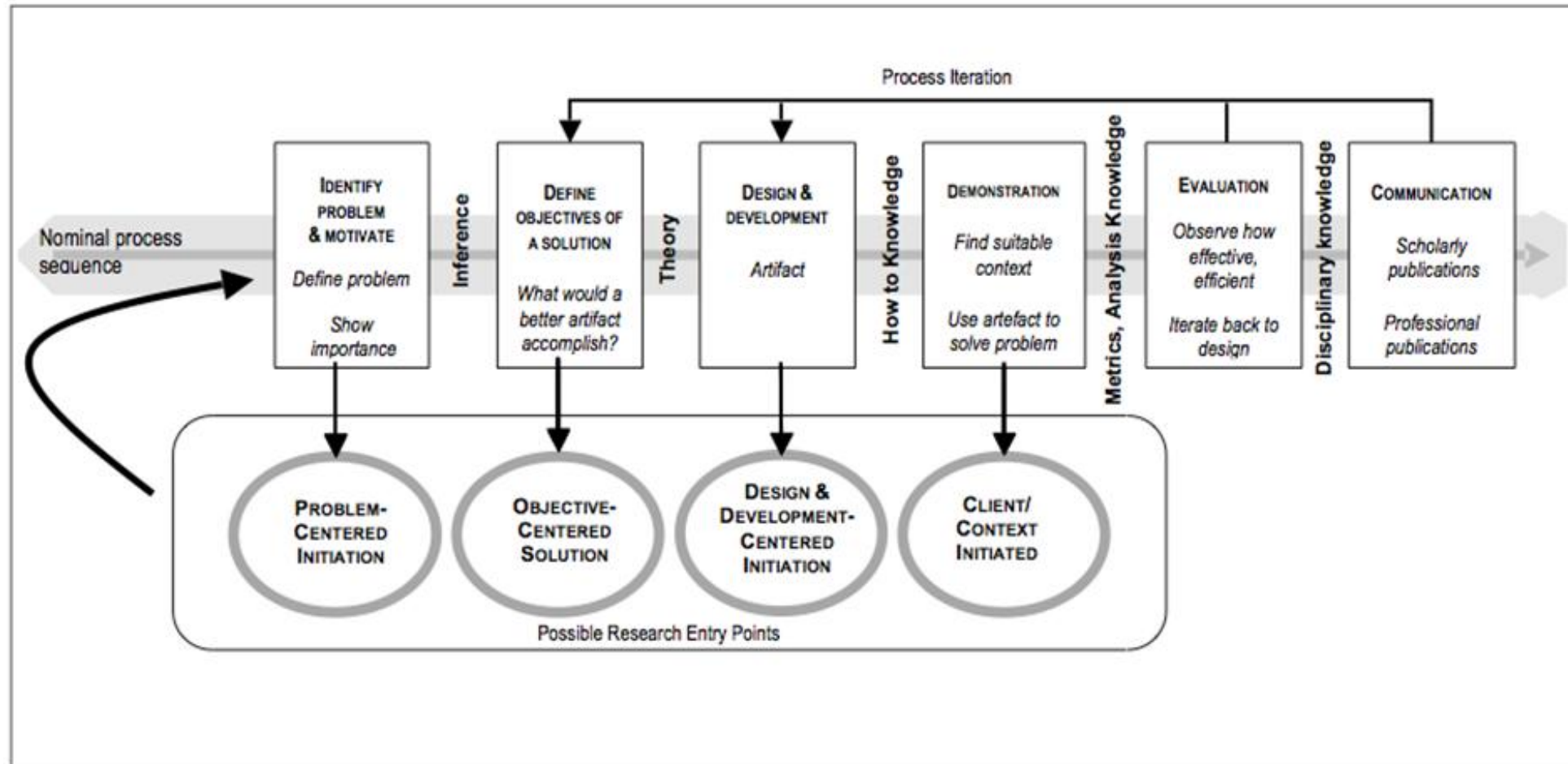
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Guideline	Description
Guideline 1: Design as an Artifact	Design-science research must produce a viable artifact in the form of a construct, a model, a method, or an instantiation.
Guideline 2: Problem Relevance	The objective of design-science research is to develop technology-based solutions to important and relevant business problems.
Guideline 3: Design Evaluation	The utility, quality, and efficacy of a design artifact must be rigorously demonstrated via well-executed evaluation methods.
Guideline 4: Research Contributions	Effective design-science research must provide clear and verifiable contributions in the areas of the design artifact, design foundations, and/or design methodologies.
Guideline 5: Research Rigor	Design-science research relies upon the application of rigorous methods in both the construction and evaluation of the design artifact.
Guideline 6: Design as a Search Process	The search for an effective artifact requires utilizing available means to reach desired ends while satisfying laws in the problem environment.
Guideline 7: Communication of Research	Design-science research must be presented effectively both to technology-oriented as well as management-oriented audiences.

DSR Methodology Process Model



From (Peppers et al., 2007-8)

DSR Methodology Process Model

- Composed by 6 activities:
 - ☐ **Problem Identification**
 - ☐ **Define Objectives for a solution**
 - ☐ **Design and Development**
 - ☐ **Demonstration**
 - ☐ **Evaluation**
 - ☐ **Communication**

DSR Methodology Process Model

Activity	Description
1- Problem identification and motivation	Define the specific research problem and justify the value of a solution. The problem definition will be used to develop an artefact that can effectively provide a solution. Resources required for this activity include knowledge of the state of the problem and the importance of its solution.
2- Define the objectives for a solution	Infer the objectives of a solution from the problem definition and knowledge of what is possible and feasible. Objectives can be quantitative (e.g., the terms in which a desirable solution would be better than current ones) or qualitative (e.g., a description of how a new artefact is expected to support solutions to problems not hitherto addressed).
3- Design and development	This activity includes determining the artefact's desired functionality and its architecture and then creating the actual artefact (e.g., constructs models, methods, or instantiations). Resources required moving from objectives to design and development include knowledge of theory that can be brought to bear in a solution
4- Demonstration	Demonstrate the use of the artefact to solve one or more instances of the problem. This could involve its use in experimentation, simulation, case study, proof, or other appropriate activity.
5- Evaluation	Observe and measure how well the artefact supports a solution to the problem. This activity involves comparing the objectives of a solution to actual observed results from use of the artefact in the demonstration. Evaluation can take many forms: a comparison of the artefact's functionality with the solution objectives (activity 2), quantitative performance measures such as the results of satisfaction surveys, client feedback, or simulations.
6- Communication	Communicate the problem and its importance. Communicate the artefact: its utility and novelty, the rigour of its design, and its effectiveness to researchers and other relevant audiences, such as practicing professionals, when appropriate. The structure of the DSRM process is also suitable to structure academic research papers, in line with the common structure for empirical research papers (problem definition, literature review, hypothesis development, data collection, analysis, results, discussion, and conclusion).

Applications of DSRM

- 4 Approaches:

- ❑ **problem-centred approach** - The basis of the nominal sequence, starting with activity 1. This line of research is adjusted for a situation *“when the idea for the research resulted from observation of the problem or from suggested future research in a paper from a prior project,”* (Peppers et al., 2007-8, p.14) ;
- ❑ **objective-centred solution** - Entry point at activity 2, and may be triggered by an industry or research need that can be addressed by developing an artefact;
- ❑ **design and development-centred approach** - Starts with activity 3, and is suitable for a situation where the artefact already exists but is not totally developed as a solution for the explicit problem domain in which it is supposed to be used;
- ❑ **client/context initiated solution**- May be based on observing a practical solution that worked, or a result of a consulting experience. In this case, the DSRM process would start at activity 4 and researchers need to go back to previous activities to apply rigour to the research.

Design Evaluation Methods

Table 2. Design Evaluation Methods	
1. Observational	Case Study: Study artifact in depth in business environment
	Field Study: Monitor use of artifact in multiple projects
2. Analytical	Static Analysis: Examine structure of artifact for static qualities (e.g., complexity)
	Architecture Analysis: Study fit of artifact into technical IS architecture
	Optimization: Demonstrate inherent optimal properties of artifact or provide optimality bounds on artifact behavior
	Dynamic Analysis: Study artifact in use for dynamic qualities (e.g., performance)
3. Experimental	Controlled Experiment: Study artifact in controlled environment for qualities (e.g., usability)
	Simulation – Execute artifact with artificial data
4. Testing	Functional (Black Box) Testing: Execute artifact interfaces to discover failures and identify defects
	Structural (White Box) Testing: Perform coverage testing of some metric (e.g., execution paths) in the artifact implementation
5. Descriptive	Informed Argument: Use information from the knowledge base (e.g., relevant research) to build a convincing argument for the artifact's utility
	Scenarios: Construct detailed scenarios around the artifact to demonstrate its utility

Evaluation

- **Evaluation of *constructs*** tends to involve completeness, simplicity, elegance, understandability, and ease of use;
- ***Models* are evaluated** in terms of their fidelity with real world phenomena, completeness, level of detail, robustness, and internal consistency;
- **Evaluation of *methods*** concerns operationality (i.e., the ability to perform the intended task or the ability of humans to effectively use the method if it is algorithmic), efficiency, generality, and ease of use; and the;
- **Evaluation of *instantiations*** concerns the efficiency and effectiveness of the artefact and its impacts on the specific environment and its users.

From (March & Smith, 1995)



References

- Slides produced from:
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